

Lenovo ThinkSystem SR650 features and specifications

Technical specifications

Lenovo

SR650 specifications

Description	Specification
System model	SR650
Form factor/height	2U
Number of processors	Up to two
Memory	Up to 1.5 TB 2666 MHz TruDDR4 Memory, 24 DIMM slots, supporting 8/16/32/64 GB DIMMs (3TB w/ 128 GB available later)
Expansion slots	One dedicated RAID slot (Internal) With one CPU: up to 4x PCIe 3.0 slots With two CPUs: up to 6x PCIe 3.0 slots
Network interface	Modular LOM support two or four 1GbE Base-T; two or four 10 GbE with Base-T or SFP+
RAID support	Support RAID 0, 1, 10, 5, 50, 6, 60, and JBOD mode with Flash or Cache support
Security	TPM 1.2 and TPM2.0 For China: TCM and TPM2.0** for China
Systems management	Lenovo XClarity Controller (Standard, Advanced, Enterprise) Lenovo XClarity Administrator / Pro / Integrators Lenovo XClarity Energy Manager Lenovo XClarity Provisioning Manager Lenovo XClarity Essentials Lenovo Capacity Planner

Description	Specification
Operating systems supported	Windows Server 2012 R2 Windows Server 2016 SUSE Linux ES 12 SP2 / 11 SP4 Red Hat RHEL 7.3 / 6.9 VMware ESXi 6.5 / 6.0 U3
Limited warranty	One or three years
Drive bays	Up to fourteen 3.5-inch bays Up to twenty-four 2.5-inch bays Option Internal M.2 module
Maximum internal storage	Up to 140 TB with 3.5-inch configuration Up to 182 TB with 2.5-inch configuration
Power supplies	Hot swap redundant 550 W, 750 W, 1100 W, 1600 W Platinum; 750 W Titanium
Cooling	Six* Hot-swap fans
Hot-swap components	Power supplies

*with configuration limitations

**available in 2H'17

6137 CPU Tcase and DTS thermal specifications

The 6137 CPU Tcase thermal based specifications are used for heatsink sizing while Digital Thermal Sensor (DTS) based specifications are used for acoustic and fan speed optimizations. The DTS reports a relative die temperature as an offset from TCC activation temperature. SKUs may share Tcase thermal profiles but they will have separate DTS based thermal profiles. Based on the Intel specification, the following applies:

- Fan speed optimization follows the DTS specification.
- Supports up to 30°C (86°F) ambient temperature under normal fan operation.

Fan temperature	CPU	DTS temperature	Fan duty
30°C (86°F) Normal	0	90.7°C (195.2°F)	100%
	1	85.2°C (185.3°F)	90%
25°C (77°F) Normal	0	87°C (188.6°F)	90%
	1	81.8°C (179.2°F)	80%
30 °C (86°F) Fan failure with power capping	0	62.7°C (144.8°F)	90%
	1	55.7°C (132.2°F)	90%
25°C (77°F) Fan failure with power capping	0	66°C (150.8°F)	80%
	1	58.5°C (137.3°F)	74%
21°C (69.8°F) Fan failure with power capping	0	70.1°C (158.1°F)	80%
	1	56.5°C (133.7°F)	74%

Note: Redundant cooling is not supported. System power capping will protect the server from random shutdown, increased failure rate, and hardware damages.

SR650 Processor thermal specifications

Parameter	Value	Note
Tcase	$T_{case} = [0.068 \times P] + 44^{\circ}\text{C} \text{ (111.2 }^{\circ}\text{F)}$	Tcase maximum = 58°C (136.4°F) at 205 W
Digital thermal sensor	$\text{TDTTS} = [0.268 \times P] + 44^{\circ}\text{C} \text{ (111.2 }^{\circ}\text{F)}$	DTS maximum = 99°C (210.2°F) at 205 W

SR650 6137 CPU Support matrix

- 8x 2.5-inch SAS/SATA/Anybay drives installed in the middle cage, leave the left and right cage empty (no backplane)
- NVMe support up to 4 with Anybay
- Rear PCIe support from Tier 1-3
- Does not support a PCIe add-in card SSD
- Does not support a graphics processing unit (GPU)
- Does not support the rear HDD
- Provides support up to 30 °C (86 °F) ambient temperature under normal fan operation
- Does not support fan redundancy